MILITARY SPECIFICATION

PRAZOSIN HYDROCHLORIDE CAPSULES

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Prazosin Hydrochloride Capsules.

1.2 Classification. The capsules covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - 1 mg
Strength 2 - 2 mg
Strength 3 - 5 mg

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-285 - Capsules (for Medicinal Purposes).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Prazosin Hydrochloride Capsules, and shall be in accordance with all applicable requirements of FED-STD-285, and as specified herein.

S2. Classification. Shall be Type I or II.

S5.4 The following is added to this paragraph:

Strength 1. Capsules shall contain prazosin hydrochloride equivalent to 1 mg of prazosin base within the designated assay limits for the capsules.

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Strength 2. Capsules shall contain prazosin hydrochloride equivalent to 2 mg of prazosin base within the designated assay limits for the capsules.

Strength 3. Capsules shall contain prazosin hydrochloride equivalent to 5 mg of prazosin base within the designated assay limits for the capsules.

The capsules shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of prazosin when determined as specified in 4.5.1.

The capsules shall comply with the requirements in the USP content uniformity test for capsules.

The prazosin in the finished capsules shall have its major band conform to the major band of the reference standard in Rf, size, and color when tested as specified in 4.5.2.

The capsules shall have a dissolution rate of not less than 50 percent in 30 minutes when determined as specified in 4.5.3.

3.2 Ingredients.

3.2.1 Prazosin hydrochloride. The prazosin hydrochloride used in the manufacture of the capsules shall comply with the following:

3.2.1.1 Identification. The prazosin hydrochloride used in the manufacture of the capsules shall be identified as specified in 4.5.4.1.

3.2.1.2 Heavy metals. Shall be not more than 50 ppm when determined using USP method II.

3.2.1.3 Residue on ignition. Shall be not more than 0.4 percent when determined using USP method.

3.2.1.4 Water content. Shall be not more than 0.5 percent when determined using USP Karl Fischer titrimetric method.

3.2.1.5 Assay. Shall assay to contain not less than 88.6 percent and not more than 94.0 percent of prazosin when determined as specified in 4.5.4.2.

3.3 Other ingredients. All other ingredients used in the manufacture of the capsules shall comply with S5.2 of FED-STD-285.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests.

4.5.1 Assay for prazosin in the finished capsule.

Reagent.

Methanolic 0.01N hydrochloric acid containing 30 percent water - Prepared by adding 0.85 ml hydrochloric acid to 300 ml of purified water, and then add sufficient methanol to make one liter.

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Select a sample of 30 capsules and assay ten of these capsules individually according to the following procedure:

Quantitatively transfer the contents of each capsule to individual glass stoppered 100 ml Erlenmeyer flasks and pipet 50 ml of methanolic 0.01N hydrochloric acid containing 30 percent water. Shake the resulting suspensions on a mechanical shaker for 15 minutes. Filter about a 30 ml portion of the contents of each flask through 1 inch diameter Gelman GA-3 filters (or equivalent) mounted in suitable filter holders into 50 ml glass stoppered centrifuge tubes. For strength 1 capsules, dilute a 25 ml aliquot of the clear filtrate with 0.01N methanolic hydrochloric acid containing 30 percent water to a final volume of 50 ml. For strength 2 capsules, dilute a 25 ml aliquot of the clear filtrate with 0.01N methanolic hydrochloric acid containing 30 percent water to a final volume of 100 ml. For strength 3 capsules, dilute a 10 ml aliquot of the clear filtrate with 0.01N methanolic hydrochloric acid containing 30 percent water to a final volume of 100 ml.

Dissolve an accurately weighed quantity of prazosin hydrochloride reference standard in methanolic 0.01N hydrochloric acid containing 30 percent water and dilute quantitatively and stepwise with methanolic 0.01N hydrochloric acid containing 30 percent water to obtain a solution having a known concentration of about 11 µg per ml. Concomitantly determine the absorbances of the reference standard solution and each sample solution in 1 cm cells at the wavelength of maximum absorbance at about 330 nm with a suitable spectrophotometer, using methanolic 0.01N hydrochloric acid containing 30 percent water as the blank. The average content of the 10 individual capsules shall assay within the limits specified in 3.1. In addition, the 10 individual capsules shall comply with the limits and the requirements of the USP Content Uniformity Test for Capsules.

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Calculate the quantity, in mg, of prazosin per capsule by the formula:

$$\text{weight (mg) of prazosin per capsule} = \frac{Au \times C \times DF \times 0.913}{As}$$

where

Au = Maximum absorbance of sample solution
As = Maximum absorbance of reference standard solution
C = Concentration of prazosin hydrochloride reference standard in mg/ml
DF = dilution factor for strength 1 capsule is $50 \times \frac{50}{25}$

dilution factor for strength 2 capsule is $50 \times \frac{100}{25}$

dilution factor for strength 3 capsule is $50 \times \frac{100}{10}$

0.913 = the fractional prazosin content of prazosin hydrochloride

4.5.2 Identification of prazosin in finished capsule.

Combine the contents of five capsules in a 50 ml glass stoppered centrifuge tube and add 10 ml of chloroform-methanol mixture (1:1, V/V) for strength 1 capsules, or 20 ml chloroform-methanol mixture (1:1, V/V) for strength 2 capsules, or 50 ml of chloroform-methanol mixture (1:1, V/V) for strength 3 capsules. Shake the resulting mixture on a mechanical shaker for 10 minutes and clarify the suspension by centrifuging at 1500 rpm for about 10 minutes. The concentration of prazosin is approximately 0.5 mg per ml. The clarified extract is treated by the following thin-layer chromatographic procedure.

Apparatus and Reagents.
Absorbent - Silica gel GF plate, 6 in by 8 in, 250 micron thick
Developing solvent - ethyl acetate-diethylamine (95:5, V/V).
Reference standard solution - Dissolve about 27.5 mg of prazosin hydrochloride reference standard accurately weighed in sufficient chloroform-methanol (1:1, V/V) to yield 50 ml of solution. The concentration of prazosin in this solution is approximately 0.5 mg per ml.

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Procedure.

Apply a 0.100 ml aliquot of the clarified extract of the capsule contents as a 2.5 inch streak to one-half of the origin line. Apply a 0.100 ml aliquot of the reference solution as a 2.5 inch streak to the other half of the origin line. In a suitable thin-layer chromatography chamber develop the thin-layer plate in the solvent (ethyl acetate-diethylamine, 95:5, V/V) using an ascending technique until the solvent front is within one inch of the top edge of the plate. After development, air-dry the plate and examine under a short wave ultraviolet light. Prazosin appears as a dark blue band at R_f about 0.3 on a yellow-green fluorescent background. The main band of the sample conforms to the main band of the reference standard in R_f, size and color.

4.5.3 Dissolution Rate.

Add 800 ml of simulated gastric fluid (prepared as per USP except for pepsin) into a one liter beaker and place in a constant temperature bath maintained at 37°C ± 2°C. Place one capsule in each of the six tubes of the basket assembly and affix a 40 mesh stainless steel screen over the top of the basket. When the temperature of the simulated gastric fluid reaches 37°C, immerse the basket into the beaker and activate the tablet disintegration tester. Adjust the apparatus so that the basket descends to 1.0 ± 0.1 cm from the bottom of the beaker on the downward stroke.

After 30 minutes remove 20 ml of test medium for strength 1 capsules or 10 ml of test medium for strength 2 or strength 3 capsules. Filter the solution through a Gelman GA-3 filter (or equivalent) mounted in a suitable filter holder. For strength 1 capsules, transfer a 15 ml aliquot of the clear filtrate to a 250 ml separatory funnel and dilute with 10 ml of simulated gastric fluid. For strength 2 capsules, transfer a 6 ml aliquot of the clear filtrate to a 250 ml separatory funnel and dilute with 20 ml of simulated gastric fluid. For strength 3 capsules, transfer a 5 ml aliquot of the clear filtrate to a 250 ml separatory funnel and dilute with 45 ml of simulated gastric fluid. For strength 1 and strength 2 capsules, add 5 ml of 1N sodium hydroxide to the separatory funnel and extract with 50 ml of methylene chloride by shaking for one minute. For strength 3 capsules, add 10 ml of 1 N sodium hydroxide to the separatory funnel and extract with 100 ml of methylene chloride by shaking for one minute.

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Immediately after the phases separate, transfer a 20 ml aliquot of the lower organic phase to a 250 ml separatory funnel containing 50 ml of hexanes and 10 ml of 0.1N hydrochloric acid. Shake the separatory funnel for one minute and allow the phases to separate.

Prepare a reference standard solution by dissolving an accurately weighed quantity of prazosin hydrochloride reference standard (about 50 mg) in about 10 ml of methanol, then add sufficient 0.01N hydrochloric acid to make 100 ml. Dilute quantitatively and stepwise with 0.1N hydrochloric acid to obtain a solution having a known concentration of about 3.5 µg per ml.

Concomitantly determine the absorbances of the reference standard solution and the sample (lower aqueous phase) in 2 cm cells at the wavelength of maximum absorbance at about 246 nm with a suitable spectrophotometer using 0.1N hydrochloric acid as the blank. Calculate the percent of prazosin dissolved in 30 minutes.

Calculation:

$$\text{percent prazosin dissolved} = \frac{Au \times C \times DF \times 800 \times 100 \times 0.913}{As \qquad\qquad LC}$$

where:

Au = Maximum absorbance of sample solution
As = Maximum absorbance of reference standard solution
C = Concentration of prazosin hydrochloride reference standard in mg/ml
DF = dilution factor for strength 1 capsule is $\frac{50}{15} \times \frac{10}{20}$

dilution factor for strength 2 capsule is $\frac{50}{6} \times \frac{10}{20}$

dilution factor for strength 3 capsule is $\frac{100}{5} \times \frac{10}{20}$

LC = label amount (mg) of prazosin in six capsules
0.913 = fractional prazosin content of prazosin hydrochloride

4.5.4 Tests for prazosin hydrochloride.

4.5.4.1 Identity. Shall comply with the following identity tests.

4.5.4.1.1 Infrared spectrum. The infrared spectrum of a potassium bromide dispersion of prazosin hydrochloride sample exhibits maxima only at the same wavelengths as that of a similar preparation of prazosin hydrochloride reference standard.

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4.5.4.1.2 Ultraviolet absorption. The ultraviolet absorption spectrum of a 7.2 µg per ml solution of prazosin hydrochloride sample in methanolic 0.01N hydrochloric acid exhibits maxima at wavelengths approximately 342 nm, 329 nm, and 246 nm as that of a similar solution of prazosin hydrochloride reference standard concomitantly measured. In addition, the respective absorptivities, calculated on the dried basis, at the points of maximum absorbance at about 329 nm and 246 nm, do not differ by more than 4 percent.

4.5.4.1.3 Thin-layer chromatography. The thin-layer chromatogram of a 0.5 mg per 0.1 ml solution of prazosin hydrochloride sample in chloroform-methanol-diethylamine (50:50:5, V/V) shall exhibit a main band in R_f, size and color similar to prazosin hydrochloride reference standard similarly prepared. The sample and standard shall be treated as directed in the procedure contained in 4.5.2.

4.5.4.2 Assay.

4.5.4.2.1 Liquid chromatography.

Instrument and conditions	- An equivalent instrument and appropriate operating conditions may be used.
Instrument	- Dupont model 830 Liquid Chromatograph or other suitable liquid chromatograph equipped with an ultraviolet absorption detector.
Chromatographic column	- 0.25 meters length by 6.3 mm O.D. by 2.1 mm I.D. packed with Zorbax-SIL or equivalent.
Mobile phase	- Methanol:Water:Acetic acid (7:2:1, V/V) with Diethylamine (2 to 25 drops per liter, to have a retention time between 6 and 10 minutes).
Stationary phase	- Zorbax-SIL or equivalent.
Pressure	- 2000 psig.
Flow rate	- About 0.2 ml per minute.
Detector	- Ultraviolet absorption at 254 nm.
Sample size	- About 5 microliters.
Sensitivity	- 16 x 10^{-2}.

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Procedure.

Accurately weigh about 100 mg of prazosin hydrochloride sample into a 100 ml volumetric flask. Add about 70 ml of methanol, shake until the sample is completely dissolved and dilute to volume. Dilute quantitatively and stepwise with methanol-water (7:3 V/V) mixture to obtain a solution having a known concentration of about 30 µg per ml. Similarly prepare a prazosin hydrochloride reference standard with the same concentration as prepared for the sample. Accurately inject 5 microliters of sample into the column, record the chromatogram and measure the area of the chromatographic peak due to prazosin. In a similar manner, accurately inject an identical volume of the standard solution and measure the area of the chromatographic peak due to the reference standard. Repeat the procedure for the sample and standard and average the values of the two determinations of the area of prazosin.

Calculate the percent of prazosin by the following calculation:

$$\text{percent Prazosin} = \frac{Au \times Wref. \times 0.913 \times 100}{As \times Wsample}$$

Where:

Au	= average value of the two determinations of the area of the prazosin sample peak
As	= average value of the two determinations of the area due to the prazosin in the reference standard peak
Wref.	= weight in mg of prazosin hydrochloride reference standard
Wsample	= weight in mg of prazosin hydrochloride sample
0.913	= fractional prazosin content of prazosin hydrochloride.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 250 capsules of one strength only, as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue of one strength only shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Forty-eight (48) units of issue of one strength only shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Forty-eight (48) units of issue of one strength only shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

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NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE BELOW 86°F".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE BELOW 86°F".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-01-039-6320	PRAZOSIN HYDROCHLORIDE CAPSULES, Equivalent to 1 mg of Prazosin Base in each Capsule, 250s	Strength 1
6505-01-039-6321	PRAZOSIN HYDROCHLORIDE CAPSULES, Equivalent to 2 mg of Prazosin Base in each Capsule, 250s	Strength 2
6505-01-040-2278	PRAZOSIN HYDROCHLORIDE CAPSULES, Equivalent to 5 mg of Prazosin Base in each Capsule, 250s	Strength 3

Preparing activity:
DLA-DM

Project No. 6505-1989

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-P-37843(DLA-DM) PRAZOSIN FYDROCHLORIDE CAPSULES

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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